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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R,S) / S3 (WP) (R,S) / S1 (PT) (S,FE) Examination November 2024 (2019 Scheme)

Course Code: EET203

Course Name: MEASUREMENTS AND INSTRUMENTATION

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1	Explain the importance of calibration of measuring instruments.	(3)
2	Explain any one method of producing controlling torque for a measuring instrument?	(3)
3	What are the causes of creeping associated with induction type energy meter?	(3)
4	How the range of a wattmeter can be extended using current transformer and potential transformer?	(3)
5	With the help of diagram explain the calibration of ammeter using potentiometer.	(3)
6	Explain the working of Sphere gaps for testing high voltage.	(3)
7	Describe the working of thermistors. What are its salient features?	(3)
8	Explain the working principal of photovoltaic cells.	(3)
9	With the help of diagram explain the working of electromagnetic flow meters.	(3)
10	Describe the role of Phasor Measurement Unit.	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11(a)	What are the different methods for producing damping torque for an indicating instrument? Explain each method with the help of neat diagrams.	(7)
(b)	Explain the construction and working of PMMC instruments with the help of neat diagrams.	(7)
12(a)	Explain the construction and working of repulsion type moving iron instrument. Also derive the expression for deflecting torque of a moving iron instrument.	(10)
(b)	How the range of a voltmeter can be extended using multipliers.	(4)

Module 2

13(a) Explain the construction and working of single phase induction type energy meter with the help of neat diagram. (10)

(b) Explain any two errors that occur in electrodynamometer type wattmeter and its compensation techniques. (4)

14(a) Prove that the deflection torque produced by a wattmeter is proportional to the power consumed in the circuit. (8)

(b) Explain the power measurement in a 3-phase circuit using two wattmeter with circuit diagram for an unbalanced load condition. (6)

Module 3

15(a) Explain how low resistance can be measured with the help of neat bridge circuit? Also prove that unknown resistance is independent on contact resistance. (10)

(b) Describe loss of charge method for the measurement of insulation resistance. (4)

16(a) With the help of circuit diagram explain how unknown self-inductance is measured using Maxwell's inductance bridge. Also draw and explain the phasor diagram. (10)

(b) Explain the working of Meggar with the help of neat diagram. (4)

Module 4

17(a) With the help of neat diagram explain the measurement of iron losses in a magnetic material using Llyod-Fisher square wattmeter method. (8)

(b) Explain the measurement of flux in a ring specimen using Flux meter. (6)

18(a) With the help of neat figures explain how B-H curve of magnetic specimen is measured using step by step method. (10)

(b) Explain how temperature can be measured with a thermocouple. (4)

Module 5

19(a) With the help of diagrams explain the construction and working of LVDT. Lists the advantages and disadvantages of LVDT. (8)

(b) Explain how frequency and phase are measured using Lissajous pattern. (6)

20(a) With the help of diagrams explain the construction and working of CRO. (8)

(b) Explain the working of strain gauge. Derive the expression for the gauge factor of a strain gauge. (6)

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